Formation of Z-Bus Matrix

Matlab Code:-

```
% Universal Z-Bus Building Algorithm (Step-by-Step Output)
clear; clc;

% Network data from the provided circuit image
% Format: [From_Bus, To_Bus, Impedance]
linedata = [
    0, 1, 1i*0.25;  % Step 1: Adds Bus 1
    1, 2, 1i*0.10;  % Step 2: Adds Bus 2
    1, 3, 1i*0.10;  % Step 3: Adds Bus 3
    2, 3, 1i*0.10;  % Step 4: Link between Bus 2 and Bus 3
    0, 2, 1i*0.25   % Step 5: Link from Bus 2 to Reference
];

% Initialize empty Zbus
Zbus = [];
nl = size(linedata, 1);

disp('=== STARTING Z-BUS ALGORITHM ===');
disp(' ');

for i = 1:nl
    fb = linedata(i, 1);
    tb = linedata(i, 2);
    z  = linedata(i, 3);

    sz = size(Zbus, 1);
```

```
% TYPE 1: New bus to reference bus
if (fb == 0 && tb > sz) || (tb == 0 && fb > sz)
    fprintf('--- STEP %d: Adding Type 1 (New Bus to Ref) ---\n', i);
    fprintf('Element: Node %d to Node %d, Z = %g + %gi\n', fb, tb, real(z), imag(z));
    if isempty(Zbus)
        Zbus = z;
    else
        Zbus = [Zbus, zeros(sz, 1); zeros(1, sz), z];
    end

% TYPE 2: New bus to an existing bus
elseif (fb > 0 && fb <= sz && tb > sz) || (tb > 0 && tb <= sz && fb > sz)
    fprintf('--- STEP %d: Adding Type 2 (New Bus to Existing) ---\n', i);
    fprintf('Element: Node %d to Node %d, Z = %g + %gi\n', fb, tb, real(z), imag(z));
    if fb <= sz
        eb = fb;
    else
        eb = tb;
    end

    new_col = Zbus(:, eb);
    new_row = Zbus(eb, :);
    new_diag = Zbus(eb, eb) + z;
    Zbus = [Zbus, new_col; new_row, new_diag];

% TYPE 3: Existing bus to reference bus
elseif (fb == 0 && tb <= sz) || (tb == 0 && fb <= sz)
    fprintf('--- STEP %d: Adding Type 3 (Existing to Ref - Kron Reduction) ---\n', i);
    fprintf('Element: Node %d to Node %d, Z = %g + %gi\n', fb, tb, real(z), imag(z));
    eb = max(fb, tb);
```

```
        new_col = Zbus(:, eb);
        new_row = Zbus(eb, :);
        new_diag = Zbus(eb, eb) + z;

        Z_temp = [Zbus, new_col; new_row, new_diag];
        Zbus = Z_temp(1:sz, 1:sz) - (Z_temp(1:sz, sz+1) * Z_temp(sz+1, 1:sz)) / Z_temp(sz+1,
sz+1);

    % TYPE 4: Existing bus to existing bus
    elseif (fb > 0 && tb > 0 && fb <= sz && tb <= sz)
        fprintf('--- STEP %d: Adding Type 4 (Existing to Existing - Kron Reduction) ---\n', i);
        fprintf('Element: Node %d to Node %d, Z = %g + %gi\n', fb, tb, real(z), imag(z));

        new_col = Zbus(:, fb) - Zbus(:, tb);
        new_row = Zbus(fb, :) - Zbus(tb, :);
        new_diag = Zbus(fb, fb) + Zbus(tb, tb) - 2 * Zbus(fb, tb) + z;

        Z_temp = [Zbus, new_col; new_row, new_diag];
        Zbus = Z_temp(1:sz, 1:sz) - (Z_temp(1:sz, sz+1) * Z_temp(sz+1, 1:sz)) / Z_temp(sz+1,
sz+1);
    end

    % THIS IS THE ADDED COMMAND TO MAKE STEPS VISIBLE
    disp('Resulting Z-Bus Matrix:');
    disp(Zbus);
    disp(' ');
end

disp('=== FINAL Z-BUS MATRIX REACHED ===');
```

PSA_EXP_2_Code.m

```matlab
% Universal Z-Bus Building Algorithm (Step-by-Step Output)
clear; clc;

% Network data from the provided circuit image
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linedata = [
    0, 1, 1i*0.25;  % Step 1: Adds Bus 1
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    1, 3, 1i*0.10;  % Step 3: Adds Bus 3
    2, 3, 1i*0.10;  % Step 4: Link between Bus 2 and Bus 3
    0, 2, 1i*0.25   % Step 5: Link from Bus 2 to Reference
];

% Initialize empty Zbus
Zbus = [];
nl = size(linedata, 1);

disp('=== STARTING Z-BUS ALGORITHM ===');
disp(' ');

for i = 1:nl
```

Command Window

```
--- STEP 5: Adding Type 3 (Existing to Ref - Kron Reduction) ---
Element: Node 0 to Node 2, Z = 0 + 0.25i
Resulting Z-Bus Matrix:
   0.0000 + 0.1397i   0.0000 + 0.1103i   0.0000 + 0.1250i
   0.0000 + 0.1103i   0.0000 + 0.1397i   0.0000 + 0.1250i
   0.0000 + 0.1250i   0.0000 + 0.1250i   0.0000 + 0.1750i


=== FINAL Z-BUS MATRIX REACHED ===
```

Result:-

=== STARTING Z-BUS ALGORITHM ===

--- STEP 1: Adding Type 1 (New Bus to Ref) ---

Element: Node 0 to Node 1, Z = 0 + 0.25i

Resulting Z-Bus Matrix:

0.0000 + 0.2500i

--- STEP 2: Adding Type 2 (New Bus to Existing) ---

Element: Node 1 to Node 2, Z = 0 + 0.1i

Resulting Z-Bus Matrix:

0.0000 + 0.2500i   0.0000 + 0.2500i

0.0000 + 0.2500i   0.0000 + 0.3500i

--- STEP 3: Adding Type 2 (New Bus to Existing) ---

Element: Node 1 to Node 3, Z = 0 + 0.1i

Resulting Z-Bus Matrix:

```
   0.0000 + 0.2500i   0.0000 + 0.2500i   0.0000 + 0.2500i
   0.0000 + 0.2500i   0.0000 + 0.3500i   0.0000 + 0.2500i
   0.0000 + 0.2500i   0.0000 + 0.2500i   0.0000 + 0.3500i
```

--- STEP 4: Adding Type 4 (Existing to Existing - Kron Reduction) ---

Element: Node 2 to Node 3, Z = 0 + 0.1i

Resulting Z-Bus Matrix:

```
   0.0000 + 0.2500i   0.0000 + 0.2500i   0.0000 + 0.2500i
   0.0000 + 0.2500i   0.0000 + 0.3167i   0.0000 + 0.2833i
   0.0000 + 0.2500i   0.0000 + 0.2833i   0.0000 + 0.3167i
```

--- STEP 5: Adding Type 3 (Existing to Ref - Kron Reduction) ---

Element: Node 0 to Node 2, Z = 0 + 0.25i

Resulting Z-Bus Matrix:

```
   0.0000 + 0.1397i   0.0000 + 0.1103i   0.0000 + 0.1250i
   0.0000 + 0.1103i   0.0000 + 0.1397i   0.0000 + 0.1250i
   0.0000 + 0.1250i   0.0000 + 0.1250i   0.0000 + 0.1750i
```

=== FINAL Z-BUS MATRIX REACHED ===